

Remote Tech Support Service – Project Documentation

1. Project Overview

The **Remote Tech Support Service** is a full-stack web-based application designed to streamline technical support ticket management.

Users can register, log in, submit support tickets, categorize issues, and track their progress.

Administrators can manage tickets, assign them to technicians, and update their statuses.

This project demonstrates secure and collaborative software engineering practices by integrating frontend, backend, and database components.

2. Team Roles and Responsibilities

Team Member	Role / Responsibilities
Abdule Touray	<i>Backend Developer</i> — Implemented server-side logic using PHP and MySQL, handling authentication, CRUD operations, and backend integration.
Rokia Touray	<i>Frontend Developer</i> — Designed the user interface using HTML, CSS (Tailwind), and JavaScript for responsiveness and user experience.
Moubarak Congacou	<i>Database Engineer</i> — Designed and implemented the MySQL database schema, relationships, and generated the EER diagram including the view <code>v_tickets_full</code> .

3. System Architecture

The project is structured using a **three-tier architecture model**:

- **Frontend (Presentation Layer):** Handles user interface through HTML, CSS (Tailwind), and JavaScript.
 - **Backend (Application Layer):** PHP scripts manage business logic, authentication, and communication between UI and database.
 - **Database (Data Layer):** MySQL database stores user accounts, tickets, comments, and configuration data with foreign key constraints.
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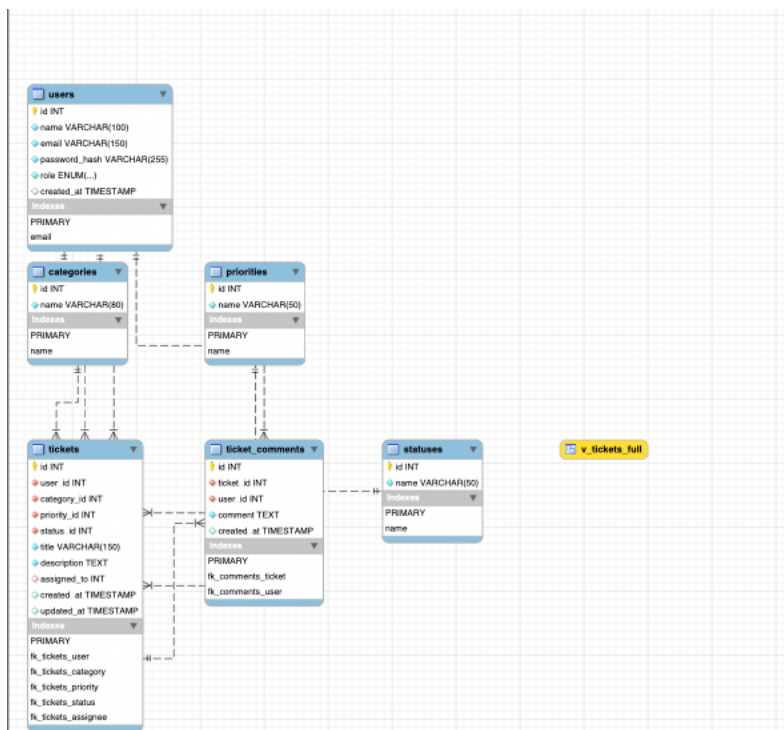
4. Database Design

The MySQL database consists of normalized tables ensuring referential integrity using foreign keys.

The schema defines users, categories, priorities, statuses, tickets, and ticket_comments, as well as a view (v_tickets_full) that aggregates ticket data for reporting and admin dashboard visualization.

5. EER Diagram

Below is the **Entity Relationship Diagram (EER)** representing all tables and their relationships.



6. Table Descriptions

Table	Description
Users	Stores user information, including names, emails, password hashes, and roles (user/admin).
categories	Contains predefined support issue categories such as Hardware, Software, and Networking.

priorities	Defines priority levels (Low, Medium, High, Critical) to classify ticket urgency.
statuses	Tracks ticket progress (Open, In Progress, Resolved, Closed).
tickets	Core table linking users, categories, priorities, and statuses with descriptions and timestamps.
ticket_comments	Records comments between users and technicians for ticket communication.
v_tickets_full	A MySQL view combining tickets, users, categories, priorities, and statuses for easy reporting.

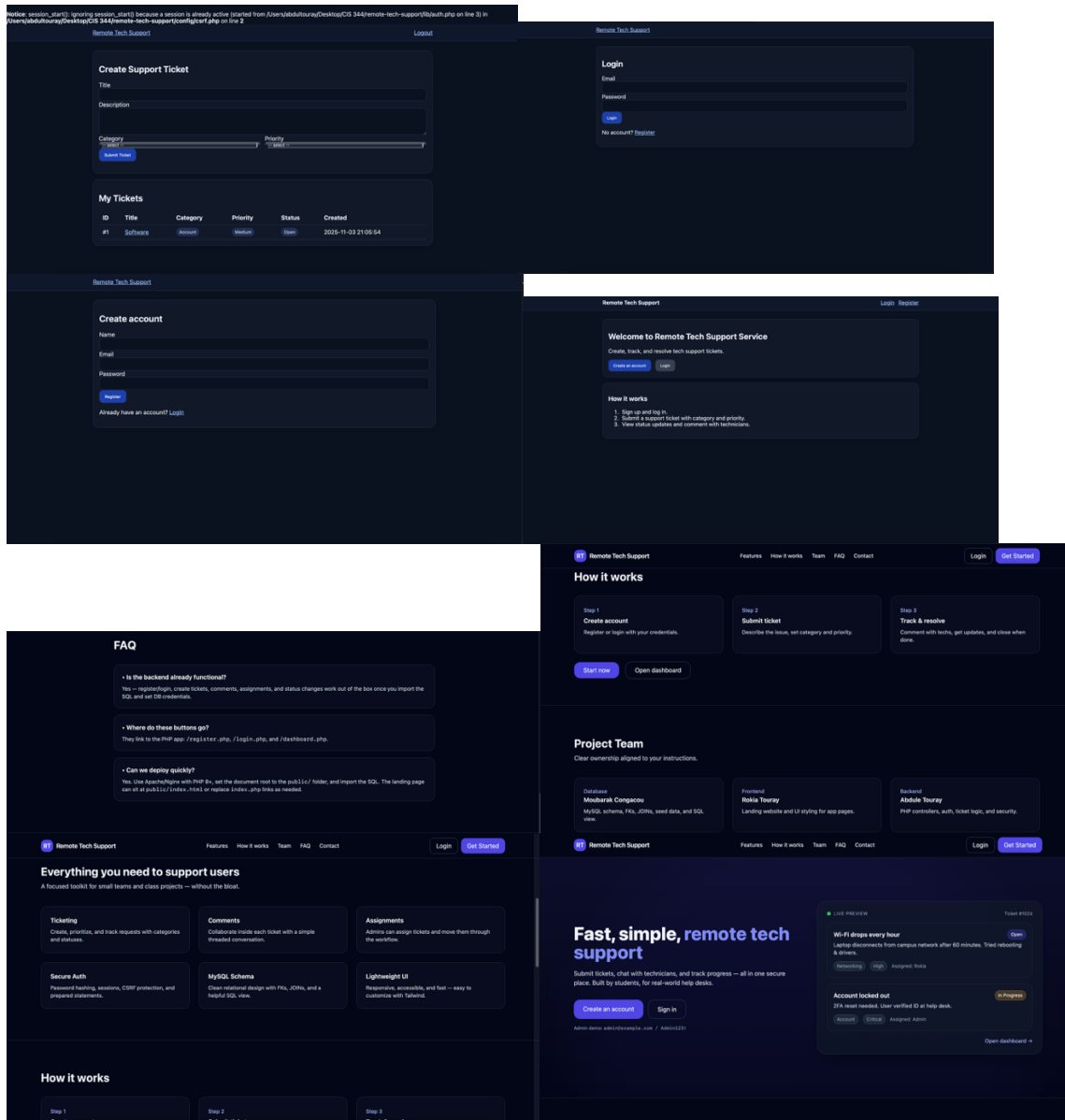
7. Setup and Configuration

1. Install **PHP (8.0 or higher)** and **MySQL (8.0 or higher)**.
2. Clone or extract the project folder to your local web server directory.
3. Create a MySQL database named `remote_support`.
4. Import the SQL scripts `schema.sql` and `sample_data.sql` from the `sql` folder.
5. Copy `config.sample.php` to `config.php` and update your MySQL credentials.
6. Run the PHP built-in server:
7. `php -S localhost:8080 -t public`
8. Open your browser and navigate to:
<http://localhost:8080>

8. Key Features

- Secure user registration and authentication (password hashing)
- Ticket creation, viewing, and management
- Admin functionality for ticket assignment and status updates
- Comment thread for communication between users and admins
- CSRF protection and sanitized inputs
- SQL view (`v_tickets_full`) for summarized ticket reporting

9. Application Screenshots (Placeholders)



10. Conclusion

The **Remote Tech Support Service** exemplifies a full-stack web development project that integrates database design, backend logic, and frontend presentation. It emphasizes modular architecture, secure coding practices, and teamwork across technical roles.

This system serves as a model for collaborative software projects using PHP, MySQL, and modern frontend design.